Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Mechatronics Engineering

Course Title	:	CAD Practice
Course Code	:	ME 2130
Experiment No	:	01
Name of Experiment	:	Introduction to Basic Solid Modeling Techniques for 3D Part Creation
Date of Experiment	:	4 July 2026
Date of Submission	:	20 July 2026

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Experiment No: 01

Name of Experiment: Introduction to Basic Solid Modeling Techniques for 3D Part Creation

1.1 Objectives

1. To become familiar with the user interface and basic functionalities of SolidWorks.
2. To learn how to create and fully define 2D sketches using various sketching and dimensioning tools.
3. To understand and apply fundamental 3D modeling features, specifically the Extruded Boss/Base and Revolved Boss/Base tools.
4. To practice creating complete 3D solid parts from given 2D drawings by following a step-by-step modeling process.

1.2 Theory

SolidWorks, by Dassault Systèmes, is a parametric Computer-Aided Design (CAD) software widely used for creating 3D solid models. The foundation of any model is a ”fully defined” 2D sketch, where geometry is precisely controlled by dimensions and geometric relations [1]. This parametric approach, starting from a robust sketch, is fundamental to creating modifiable and predictable parts [2].

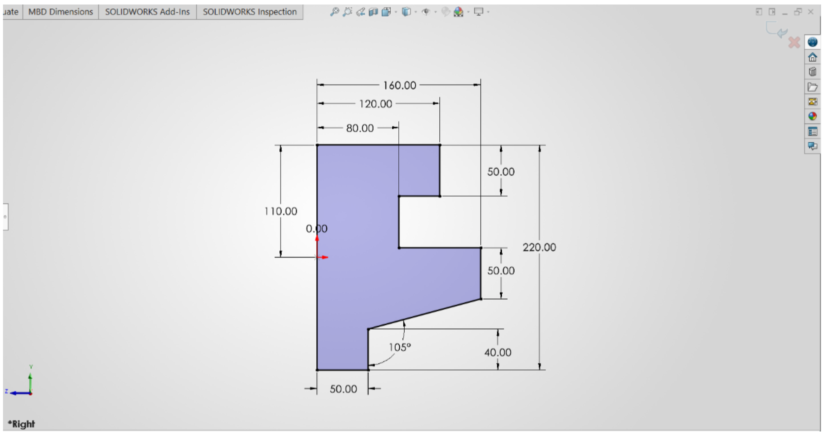


Figure 1.1: An example of a fully defined 2D sketch.

Once a 2D sketch is created, it can be transformed into a 3D part using various features. One of the most fundamental features is the Extruded Boss/Base. This tool adds depth to a selected sketch profile by projecting it linearly for a specified distance [3]. It is the primary method for creating prismatic parts and adding material based on a 2D shape [1].

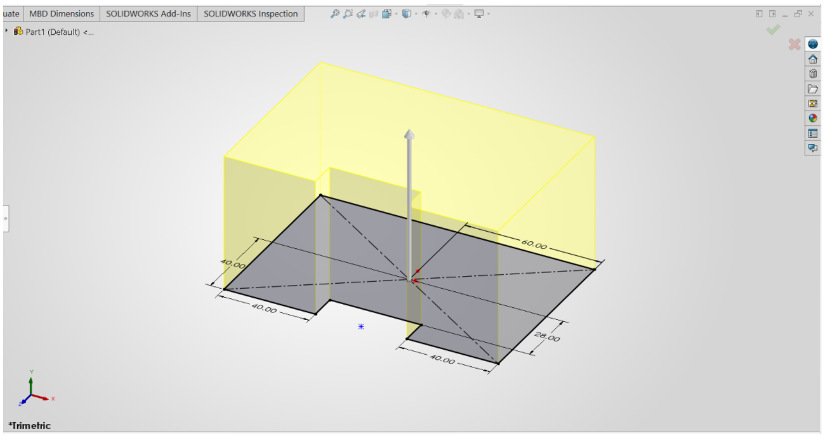


Figure 1.2: Illustration of the Extruded Boss/Base feature.

Another essential feature for creating cylindrical or axisymmetric parts is the Revolved Boss/Base. This tool creates a 3D model by revolving a 2D sketch around a specified centerline or axis [3]. It is highly efficient for modeling objects like wheels, shafts, and other parts with rotational symmetry. The ability to master both extrusion and revolution provides a powerful toolkit for modeling a vast array of mechanical components [2].

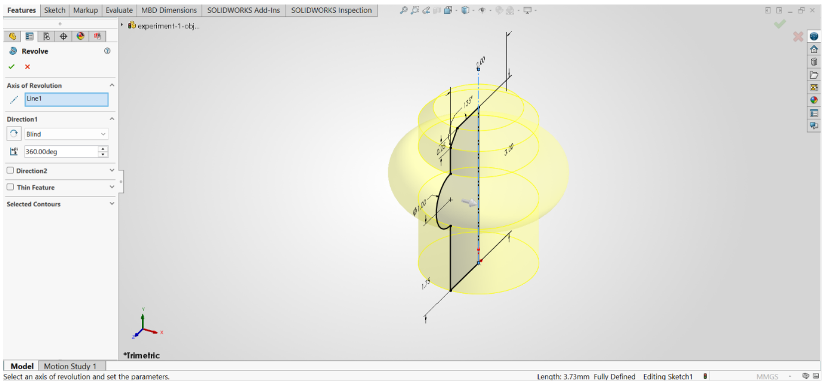


Figure 1.3: Illustration of the Revolved Boss/Base feature.

1.3 Apparatus Required

1. **Hardware:** A personal computer (PC) with hardware specifications sufficient for running SolidWorks smoothly.
2. **Software:** An installed and licensed version of the SolidWorks 3D CAD software by Dassault Systèmes.

1.4 Working

The creation of any 3D part in SolidWorks begins with a systematic workflow. This section outlines the initial steps required to set up the environment and create a basic 2D sketch, which forms the foundation for 3D features.

1.4.1 Unit System Selection

Upon launching SolidWorks and creating a new part file, the first crucial step is to define the unit system for the model. The dimensions of all sketches and features will be based on this selection. The unit system can be accessed and changed from the bottom-right corner of the status bar. For this lab, the MMGS (millimeter, gram, second) system is used for all models.

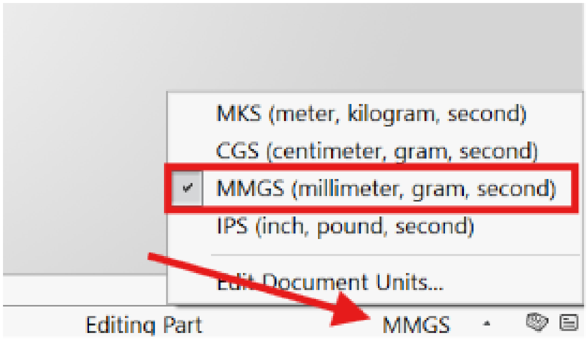


Figure 1.4: Selecting the MMGS unit system from the status bar.

1.4.2 Plane Selection

After setting the unit system, a base plane must be selected to host the initial 2D sketch. SolidWorks provides three default planes: Front, Top, and Right. The choice of plane is important as it determines the orientation of the part. To start a sketch, one must select a plane from the FeatureManager Design Tree or by clicking on it in the graphics area, and then select the Sketch command.

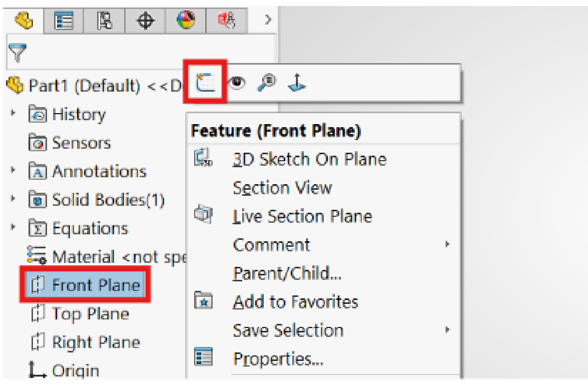
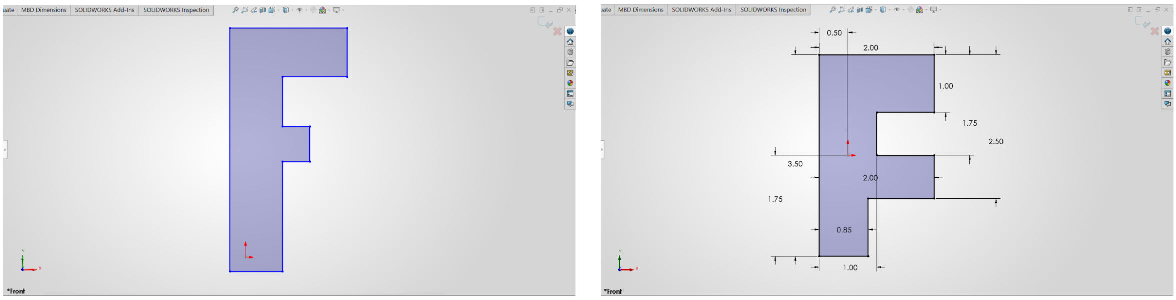


Figure 1.5: Selecting the Front Plane for the initial sketch.

1.4.3 Sketching and Dimensioning

With a plane selected and the sketch mode active, 2D geometry can be created using the tools in the Sketch CommandManager. Tools like Line, Circle, and Rectangle are used to draw the basic profile of the part.

After creating the geometry, it must be fully defined using dimensions and geometric relations. The Smart Dimension tool is used to add precise dimensional values to the sketch entities. Adding relations like Vertical, Horizontal, or Tangent helps to lock the geometry's position and orientation. A fully defined sketch is indicated by all lines turning black, and the status bar will display "Fully Defined." This ensures that the model is robust and will behave predictably when changes are made.



(a) Drawing lines and shapes for the sketch profile. **(b)** Using Smart Dimension to fully define the sketch.

Figure 1.6: Process of creating and defining a 2D sketch.

1.4.4 Extruded Boss/Base

Once a sketch is fully defined, the Extruded Boss/Base feature is used to convert the 2D profile into a 3D solid. This tool adds material by projecting the sketch perpendicular to the sketch plane for a specified depth. Various end conditions, such as *Blind*, *Through All*, and *Mid Plane*, offer control over the extrusion, making it a versatile tool for creating a wide range of prismatic parts.

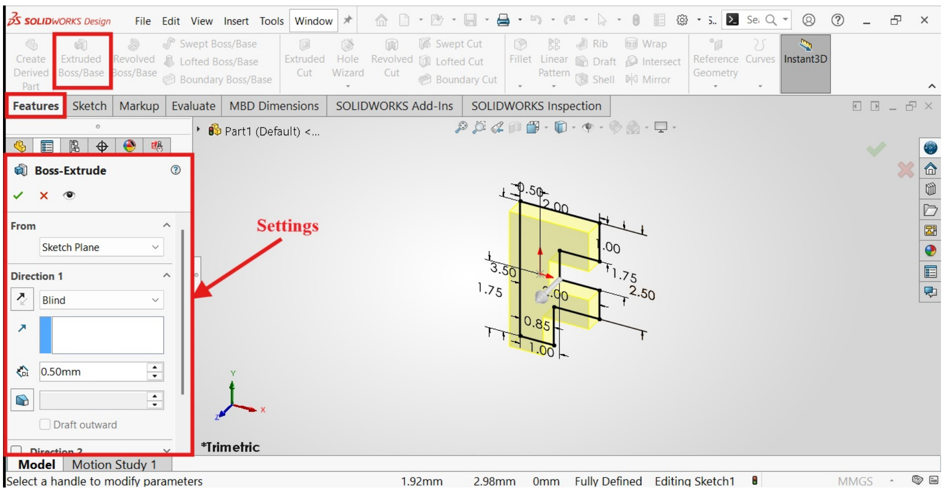


Figure 1.7: Applying the Extruded Boss/Base feature to a 2D sketch.

1.4.5 Revolved Boss/Base

For parts with cylindrical or axisymmetric geometry, the Revolved Boss/Base feature is used. This tool creates a 3D solid by revolving a 2D sketch around a selected centerline or axis. It is highly efficient for modeling objects such as wheels, shafts, and pulleys. The sketch profile defines the cross-section of the revolved part.

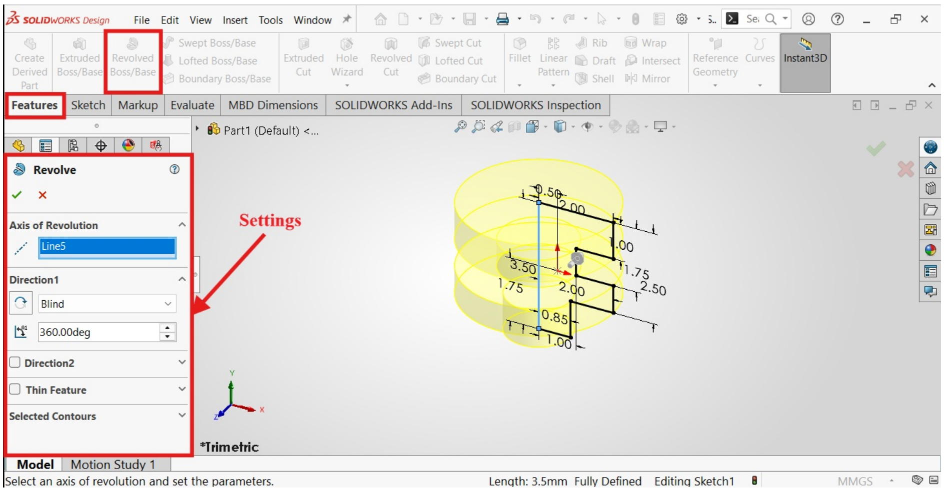


Figure 1.8: Using the Revolved Boss/Base feature with a sketch and centerline.

1.4.6 Fillet Feature

The Fillet feature is an applied feature used to create rounded internal or external faces along one or more edges. Fillets are essential for removing sharp corners, which can help reduce stress concentrations in mechanical parts and improve aesthetics. The feature allows for a constant or variable radius to be applied to the selected edges.

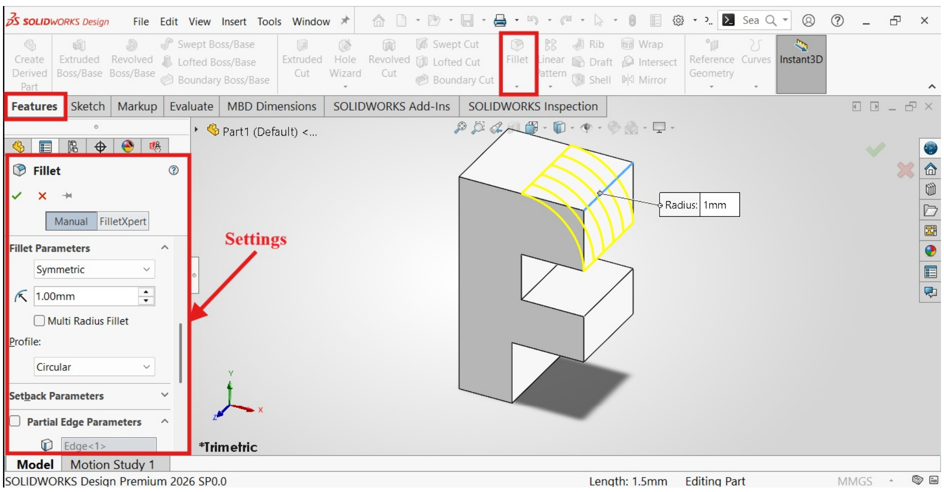
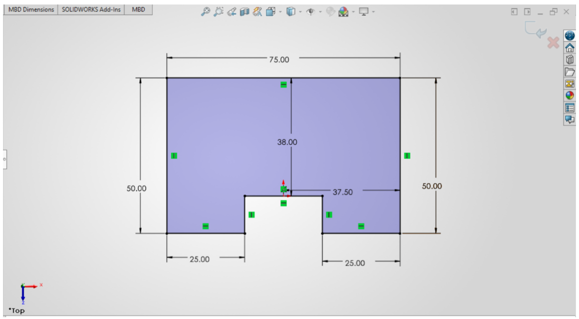


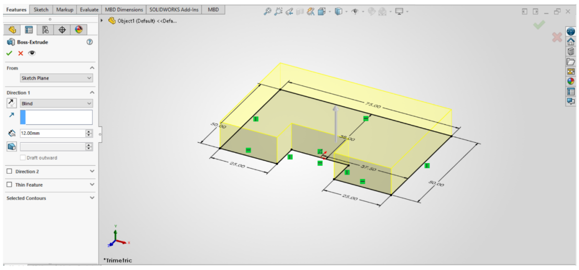
Figure 1.9: Applying a fillet to round sharp edges on a model.

1.5 Object 1

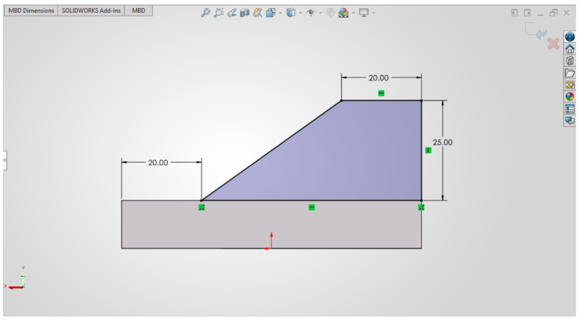
This object was modeled using a straightforward, two-part extrusion process. Initially, the base profile was created as a 2D sketch and fully defined with dimensional constraints. This sketch was then extruded to form the primary block. Subsequently, a second sketch was drawn on the top surface of the base to define the upper tier, which was also extruded to complete the final geometry.



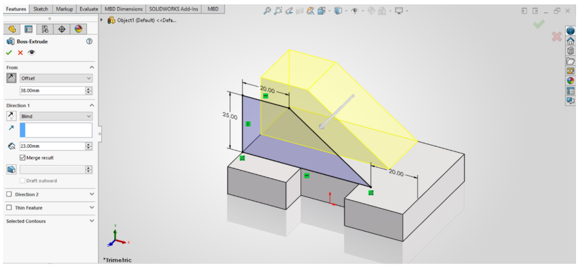
(a) 2D sketch of the base with dimensions



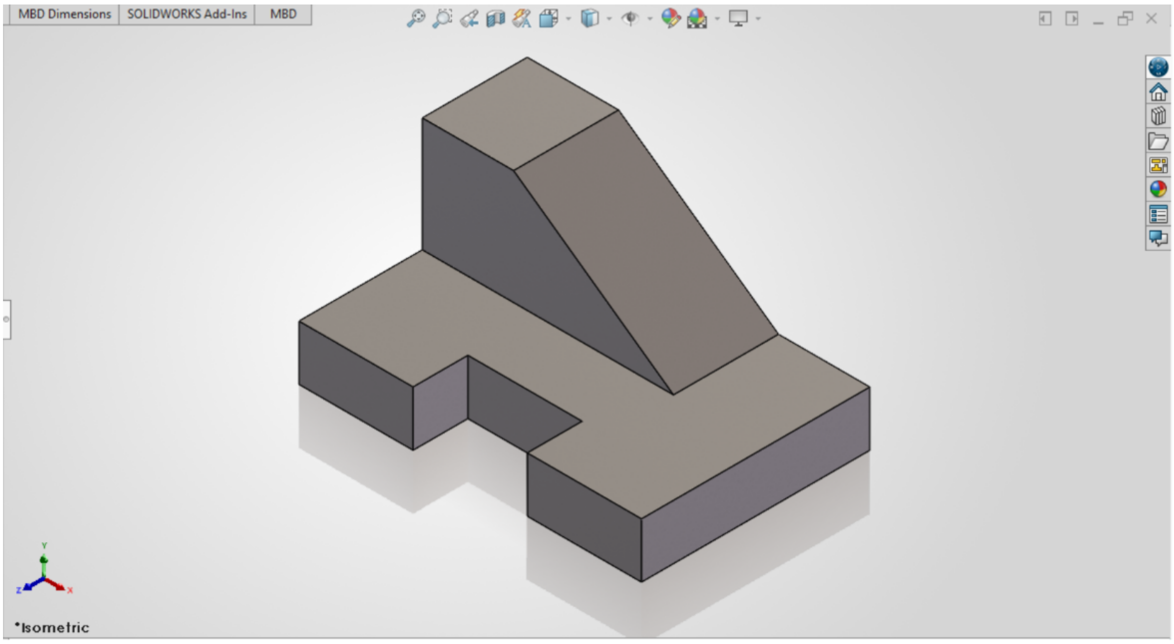
(b) Extruding the base sketch



(c) 2D sketch of the top portion with dimensions



(d) Extruding the top portion

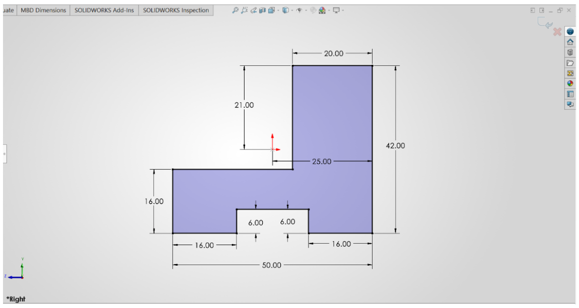


(e) Final result

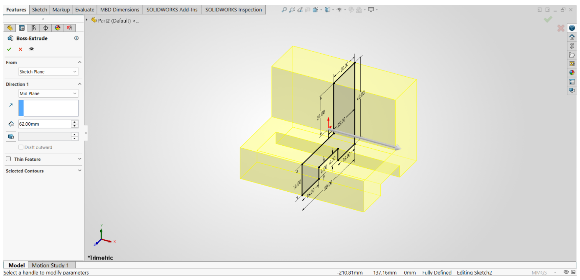
Figure 1.10: Snapshots while Designing Object 1 in Solidworks

1.6 Object 2

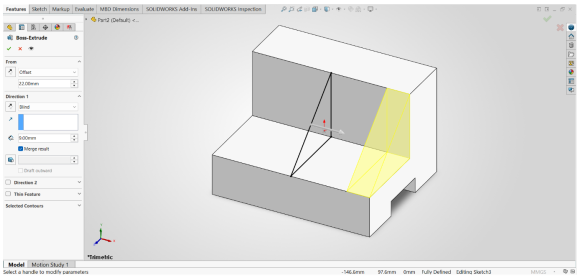
The construction of this object involved multiple features. The process began with a 2D sketch of the main body's profile, which was then extruded symmetrically using the "Mid Plane" end condition. Subsequently, a sketch for the top extension was created and extruded. The final step involved creating a sketch on the bottom face of the model and using the "Extruded Cut" feature to remove material and create the final shape.



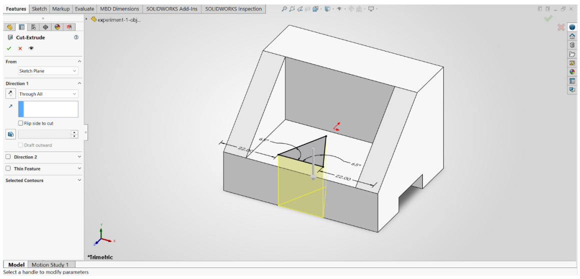
(a) 2D sketch of the main body with dimensions.



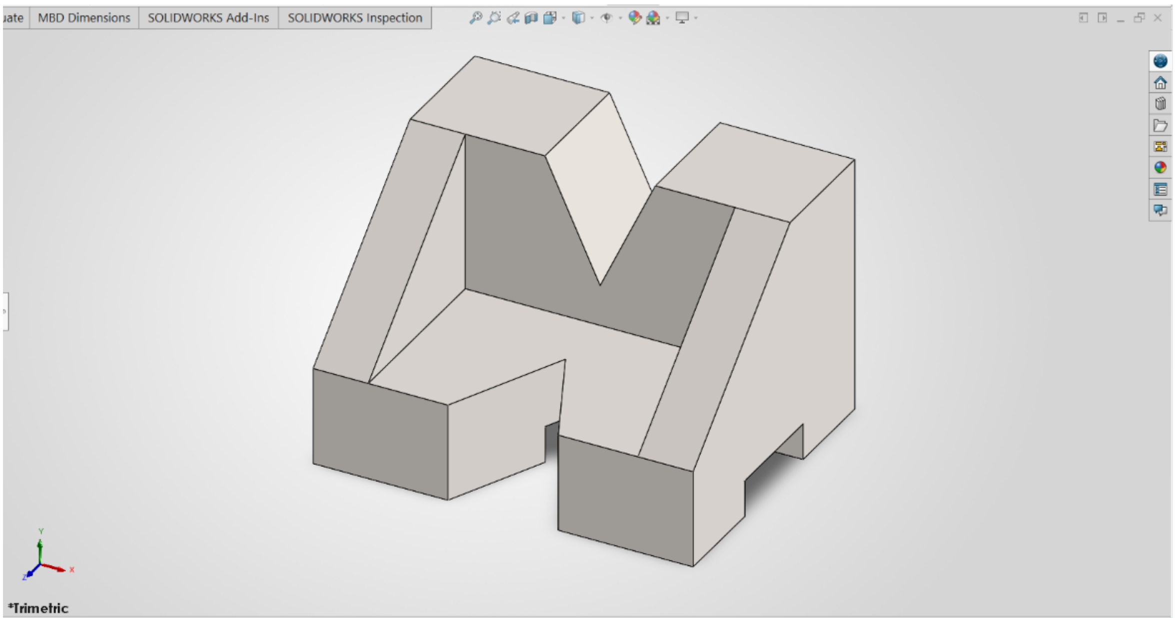
(b) Extruding the sketch from the mid-plane.



(c) Extruding the top extension.



(d) Applying the Extruded Cut feature.

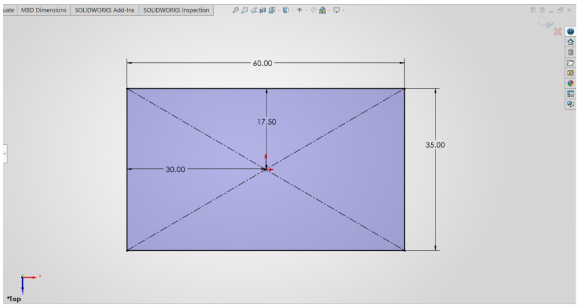


(e) Final Result.

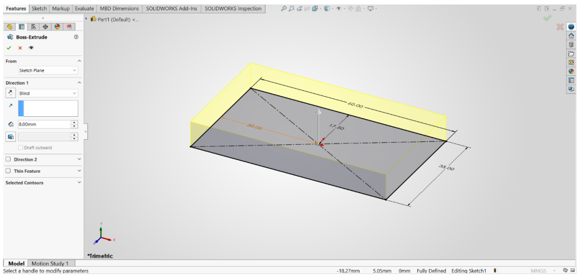
Figure 1.11: Snapshots while Designing Object 2 in Solidworks

1.7 Object 3

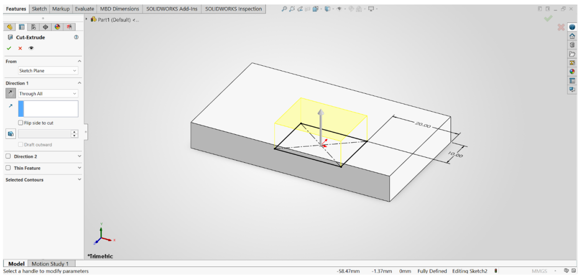
This object was created using a combination of additive and subtractive modeling techniques. The process began by sketching and extruding the main base. Following this, the "Extruded Cut" feature was utilized to remove a rectangular section from the center. Finally, two smaller rectangular profiles were sketched on the top surface and extruded upwards to form the corner posts, completing the part's geometry.



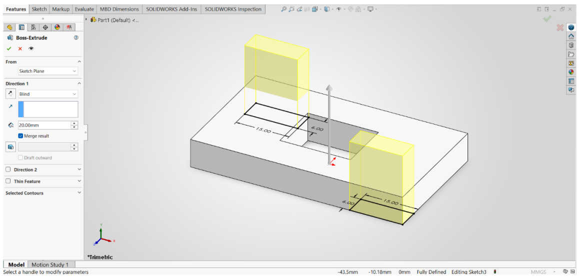
(a) 2D sketch of the base with dimensions.



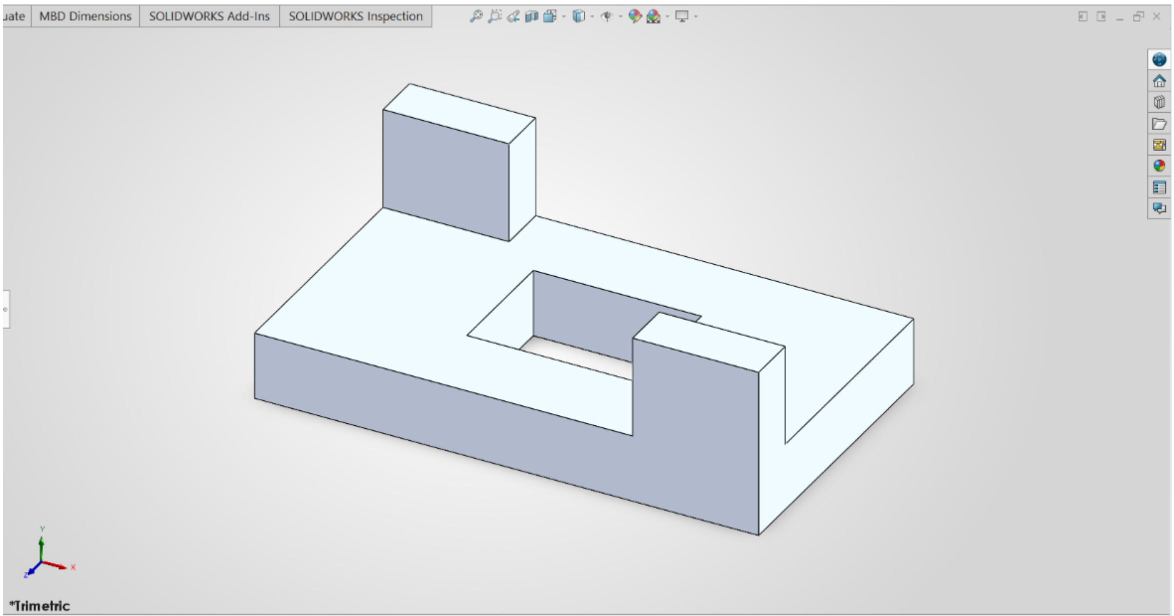
(b) Extruding the base sketch.



(c) Cutting a rectangle using the Extruded Cut feature.



(d) Extruding two rectangles from the corners.

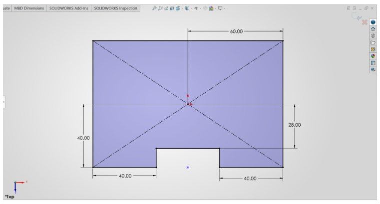


(e) Final Result.

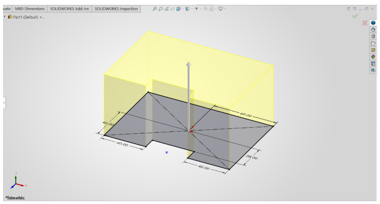
Figure 1.12: Snapshots while Designing Object 3 in Solidworks

1.8 Object 4

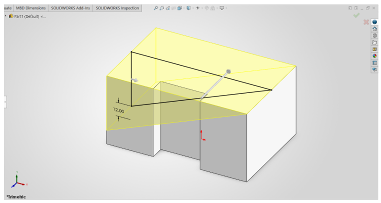
This part was constructed using a multi-step process involving both additive and subtractive methods. First, the main rectangular base was created by sketching its profile and applying an Extruded Boss/Base feature. Subsequently, a large angled cut was made using the Extruded Cut tool. A new sketch was then created on the top surface to extrude a central block upwards. The final step involved another Extruded Cut to remove a rectangular section from this newly created block, completing the model.



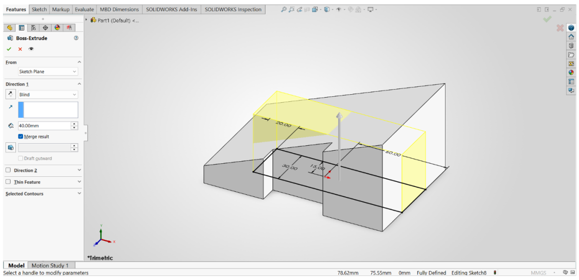
(a) 2D sketch of the base.



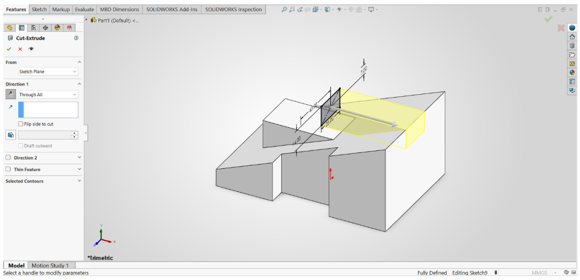
(b) Extruding the base sketch.



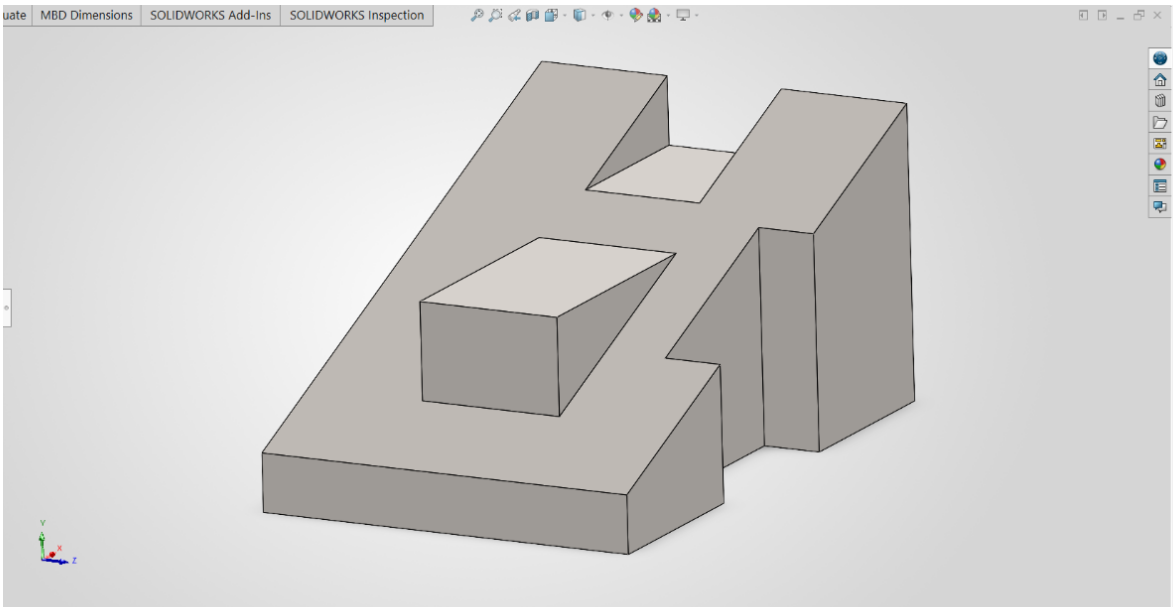
(c) Creating the angled cut.



(d) Extruding the central block.



(e) Creating the top cut-out.

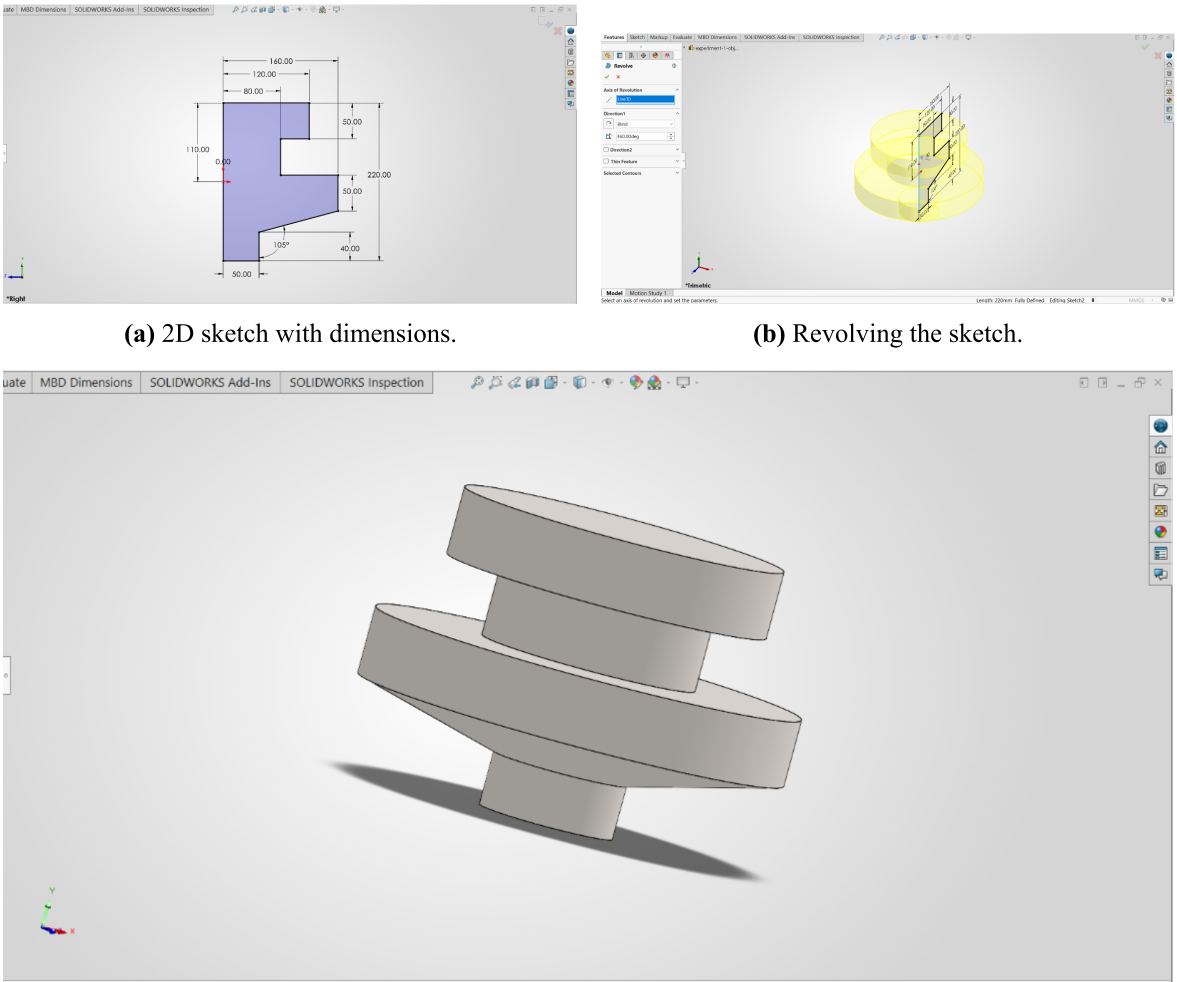


(f) Final result.

Figure 1.13: Snapshots while Designing Object 4 in Solidworks

1.9 Object 5

This object, which has rotational symmetry, was efficiently modeled using the Revolved Boss/Base feature. The process involved creating a 2D sketch of the part's cross-sectional profile and fully defining it with dimensions. The sketch was then revolved 360 degrees around a central axis to generate the final three-dimensional geometry.



(a) 2D sketch with dimensions.

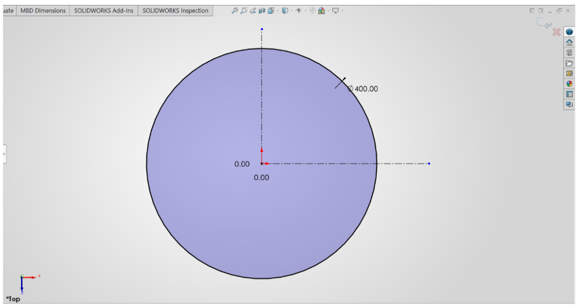
(b) Revolving the sketch.

(c) Final Result.

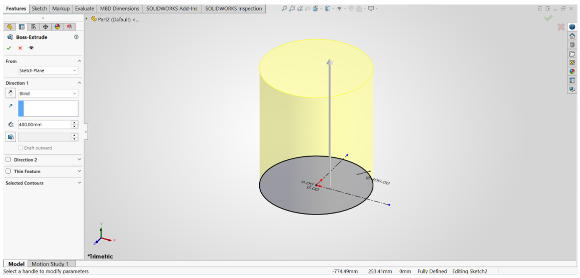
Figure 1.14: Snapshots while Designing Object 5 in Solidworks

1.10 Object 6

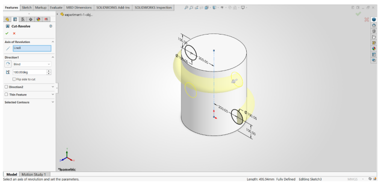
This object demonstrates the use of multiple feature types to build a complex part. The process began with creating and extruding the main base. Subsequently, a revolved feature was used to create circular geometry on two diagonal corners. To finalize the part, applied features were used: a Fillet was added to round a top edge, and a Chamfer was applied to create a beveled bottom edge, showcasing how different tools are combined in modeling.



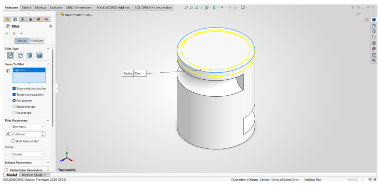
(a) 2D sketch of the base with dimensions.



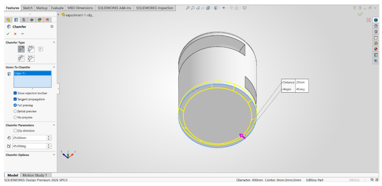
(b) Extruding the base sketch.



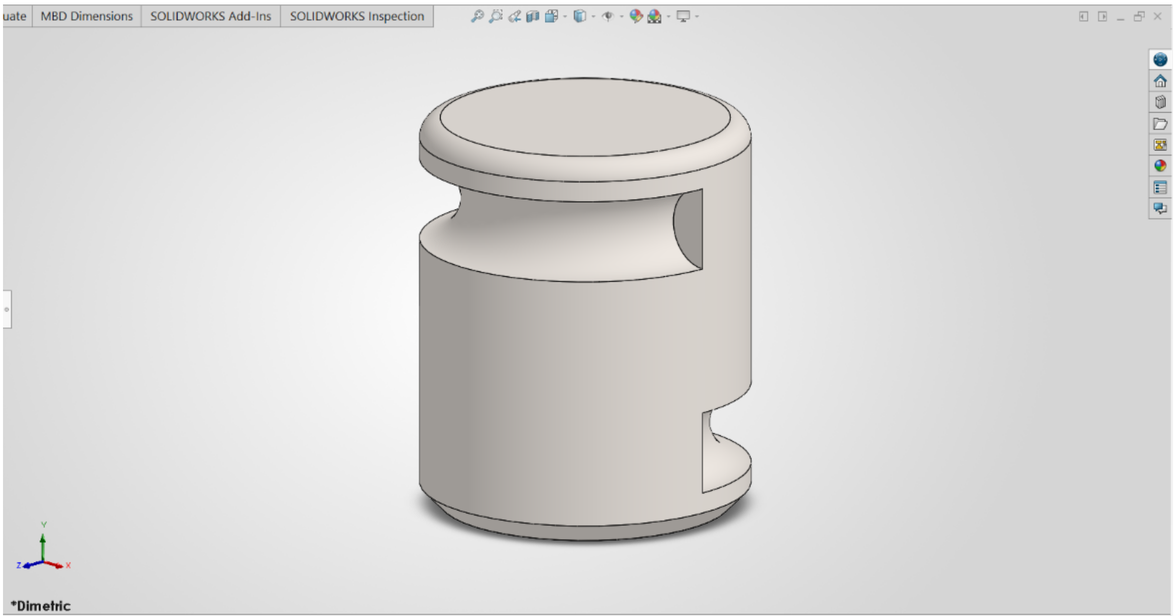
(c) Creating revolved features on diagonal corners.



(d) Applying a Fillet feature to the top edge.



(e) Applying a Chamfer feature to the bottom edge.

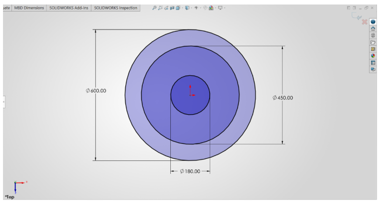


(f) Final Result.

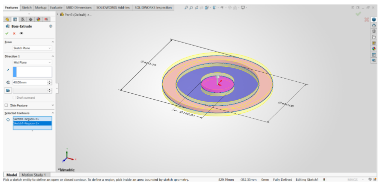
Figure 1.15: Snapshots while Designing Object 6 in Solidworks

1.11 Object 7

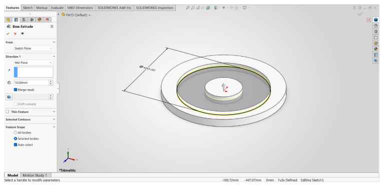
This object, featuring rotational symmetry, was modeled efficiently using the Revolved Boss/Base feature. The process involved creating a 2D sketch of the part's cross-sectional profile and fully defining it with dimensions. This sketch was then revolved 360 degrees around a central axis to generate the final three-dimensional geometry.



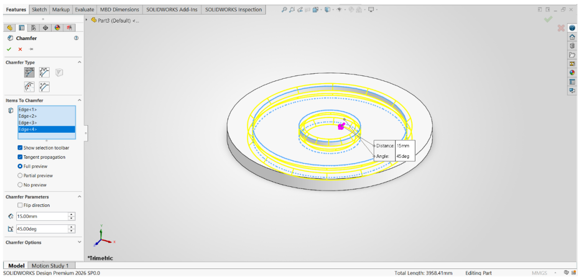
(a) 2D sketch of the base with dimensions.



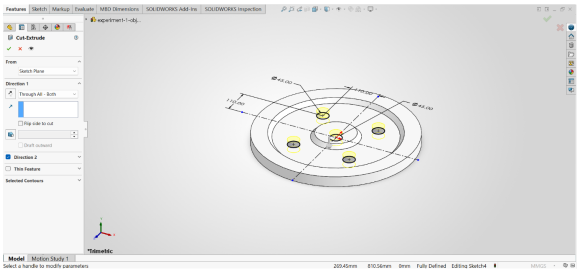
(b) Extruding the outer and middle circle.



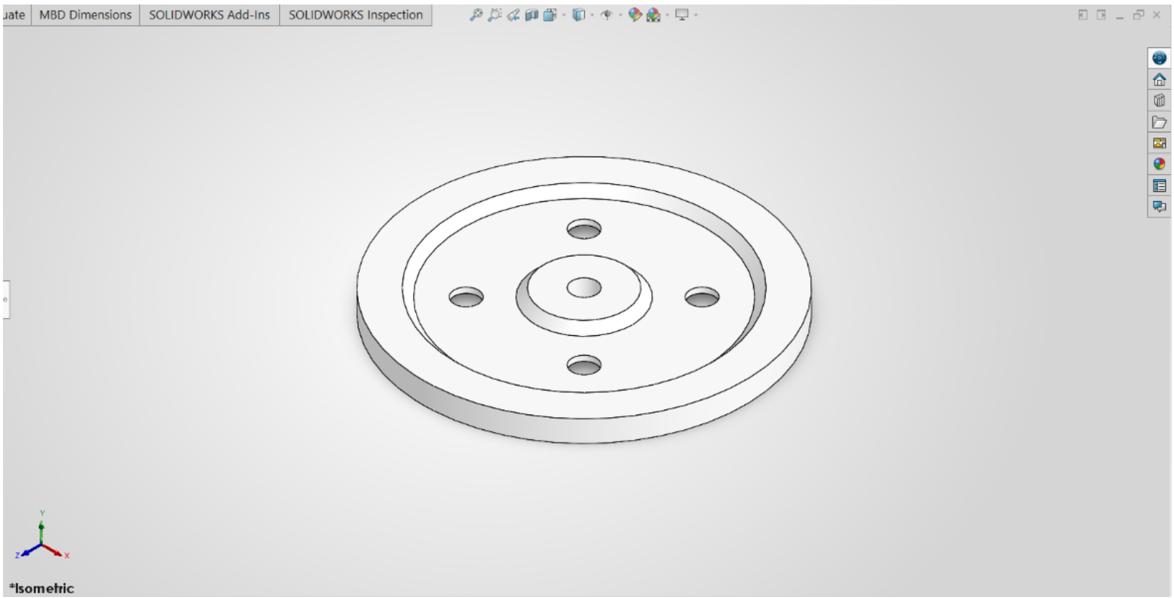
(c) Extruding the inner circle.



(d) Applying chamfer to the edges



(e) Cutting off the holes

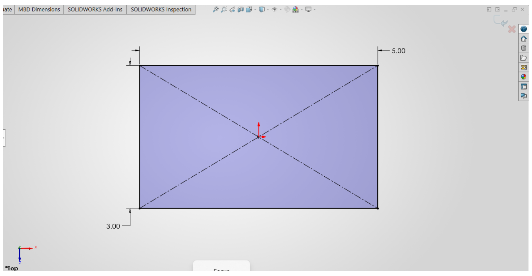


(f) Final result.

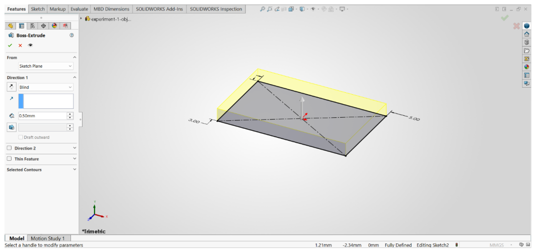
Figure 1.16: Snapshots while Designing Object 7 in Solidworks

1.12 Object 8

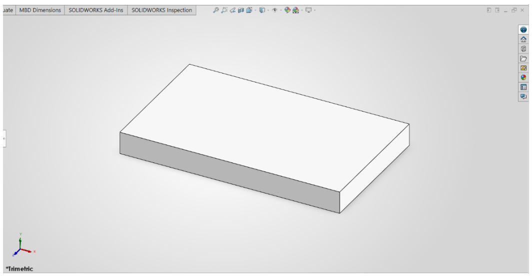
This object, which features rotational symmetry, was modeled efficiently using the Revolved Boss/Base feature. The process involved creating a 2D sketch of the part's cross-sectional profile and fully defining it with dimensions. This sketch was then revolved 360 degrees around a central axis to generate the final three-dimensional geometry.



(a) 2D sketch with dimensions



(b) Revolving the sketch around the vertical axis

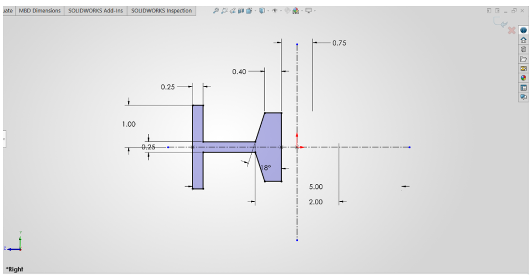


(c) Final Result

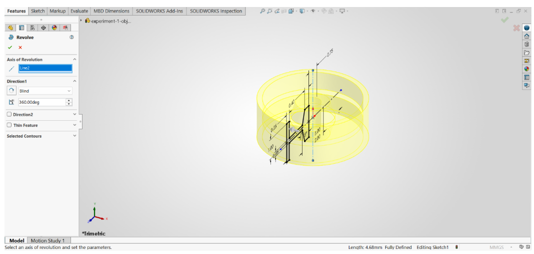
Figure 1.17: Snapshots while Designing Object 8 in Solidworks

1.13 Object 9

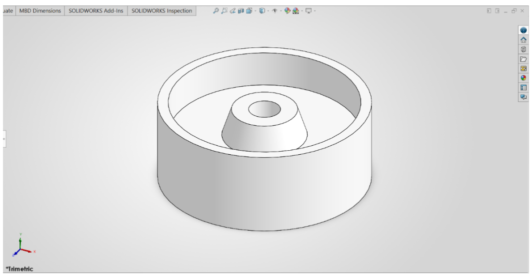
This object, which features rotational symmetry, was modeled efficiently using the Revolved Boss/Base feature. The process involved creating a 2D sketch of the part's cross-sectional profile and fully defining it with dimensions. This sketch was then revolved 360 degrees around a central axis to generate the final three-dimensional geometry.



(a) 2D sketch with dimensions



(b) Revolving the sketch around the vertical axis

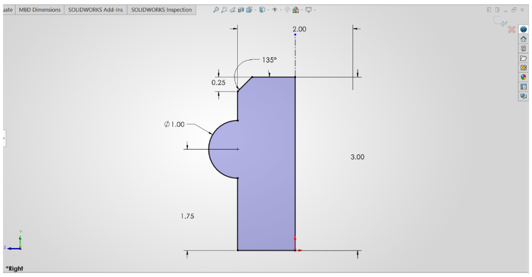


(c) Final Result

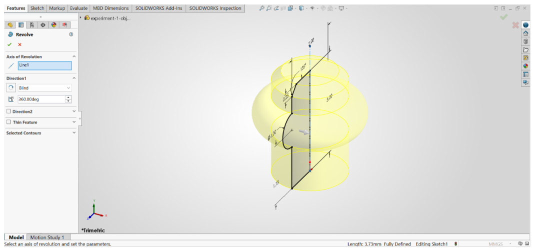
Figure 1.18: Snapshots while Designing Object 9 in Solidworks

1.14 Object 10

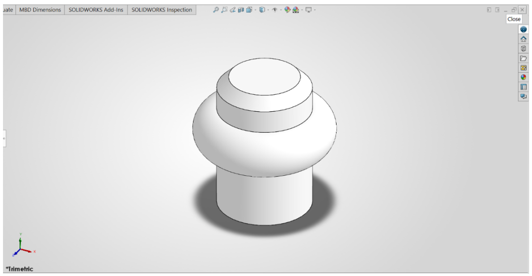
This object, which features rotational symmetry, was modeled efficiently using the Revolved Boss/Base feature. The process involved creating a 2D sketch of the part's cross-sectional profile and fully defining it with dimensions. This sketch was then revolved 360 degrees around a central axis to generate the final three-dimensional geometry.



(a) 2D sketch with dimensions



(b) Revolving the sketch around the vertical axis



(c) Final Result

Figure 1.19: Snapshots while Designing Object 10 in Solidworks

1.15 Discussion

1.16 Conclusion

References

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